

Background

The Bottleneck of OMP:

- An incorrect choice in atom selection stage propagates through subsequent iterations, degrading recovery.

Limitation of General Matrices:

- Conventional sensing matrices mainly rely on general incoherence properties.
- They fail to capture the specific energy distributions of datasets, leading to suboptimal matching.

$$\lambda = \operatorname{argmax}_j |\Psi_j^T \mathbf{r}|$$

Sensing Matrix Residual

$$\lambda = \operatorname{argmax}_j |\Psi_j^T \mathbf{r}|$$

Aim and Objectives

- Propose a data-driven sensing matrix learning framework to improve OMP's matching stage.
- To evaluate the proposed method on various datasets and compare it with existing sensing matrix methods.

Method

- Training Data Formulation:** Extract structurally similar signals data of $\mathbf{x}$ and combined all as training matrix $\mathbf{X}$.
- Optimization Problem:** Learn an equivalent sensing matrix Ψ by minimizing the proxy reconstruction error with Tikhonov regularization:

$$\min_{\Psi} \frac{1}{l} \sum_{i=1}^l \|\Psi^T \Phi \mathbf{x}_i - \mathbf{x}_i\|_2^2 + \tau \|\Psi\|_F^2$$

- Taking the derivative with respect to Ψ and setting it to zero to obtain the optimization equation, because the objective function is convex.

$$\frac{\partial J(\Psi)}{\partial \Psi} = \frac{2}{l} (\Phi \mathbf{X} \mathbf{X}^T \Phi^T) \Psi - \frac{2}{l} (\Phi \mathbf{X} \mathbf{X}^T) + 2\tau \Psi = 0.$$

- Learned Result:** The proposed equivalent sensing matrix can then be represented as:

$$\Psi = (\Phi \mathbf{X} \mathbf{X}^T \Phi^T + l\tau \mathbf{I}_m)^{-1} \Phi \mathbf{X} \mathbf{X}^T.$$

Proposed Algorithm

Algorithm 1 Proposed Sensing Matrix Design

Input: Measurement matrix Φ , regularization parameter τ , structurally similar signal set $\mathbf{X}$.

Output: $\Psi = (\Phi \mathbf{X} \mathbf{X}^T \Phi^T + l\tau \mathbf{I}_m)^{-1} \Phi \mathbf{X} \mathbf{X}^T$.

Algorithm 2 OMP via Learned Ψ

Input: Observation $\mathbf{y}$, sensing matrix Ψ , measurement matrix Φ , sparsity k .

Output: Estimated signal $\hat{\mathbf{s}}$, support set Λ .

- Initialization:** $\mathbf{r} = \mathbf{y}$, $\Lambda = \emptyset$.
- Repeat k times:
 - $\lambda = \operatorname{argmax}_j |\Psi_j^T \mathbf{r}|$,
 - $\Lambda = \Lambda \cup \{\lambda\}$,
 - $\hat{\mathbf{s}}_{\Lambda} = (\Phi_{\Lambda}^T \Phi_{\Lambda})^{-1} \Phi_{\Lambda}^T \mathbf{y}$,
 - $\mathbf{r} = \mathbf{y} - \Phi_{\Lambda} \hat{\mathbf{s}}_{\Lambda}$,
 - $\mathbf{s}_{\hat{\Lambda}} = \mathbf{0}$, $\mathbf{s}_{\Lambda} = \hat{\mathbf{s}}_{\Lambda}$.

Reference

- Schnass, K. and Vandergheynst, P. (2008) 'Dictionary preconditioning for greedy algorithms', Signal Processing, 56(5), pp. 1994–2002.
- Bo Li and Shuai Zhang, (2025) 'Robust Sensing Dictionary Design for the OMP Algorithm in Compressive Sensing,' Signal Processing, vol. 227.

Enhancement Mechanism

Gram Matrix Heatmap: Inner products between the sensing matrix and the measurement matrix columns. • **Reduced Interference:** Lower off-diagonal mean (-0.009 vs. Baseline) and sharper diagonals indicate effectively suppressed cross-correlations. • **Structural Focus:** Vertical stripes highlight the data-driven emphasis on structurally relevant atoms.

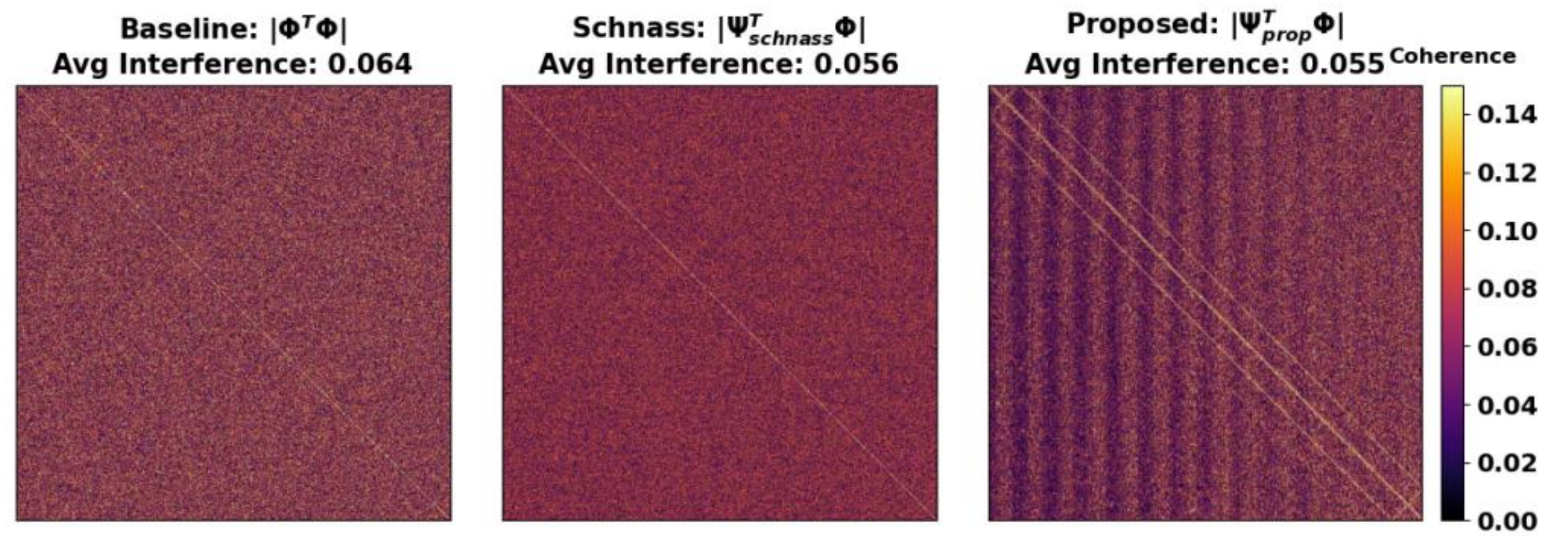


Fig. 1 The heatmaps of the Gram matrix when the CR is 0.2 and the sparsity level is 30, with a learning length size of 100.

Histogram of Effective Interference: Probability density of inner products between true support and non-target candidate atoms. • **Outlier Suppression:** Mitigates extreme values that trigger incorrect atom selection. • **Performance Improvement:** Achieves a lower mean (-0.042 vs. Schnass) and lower maximum interference (-0.0492 vs. Baseline).

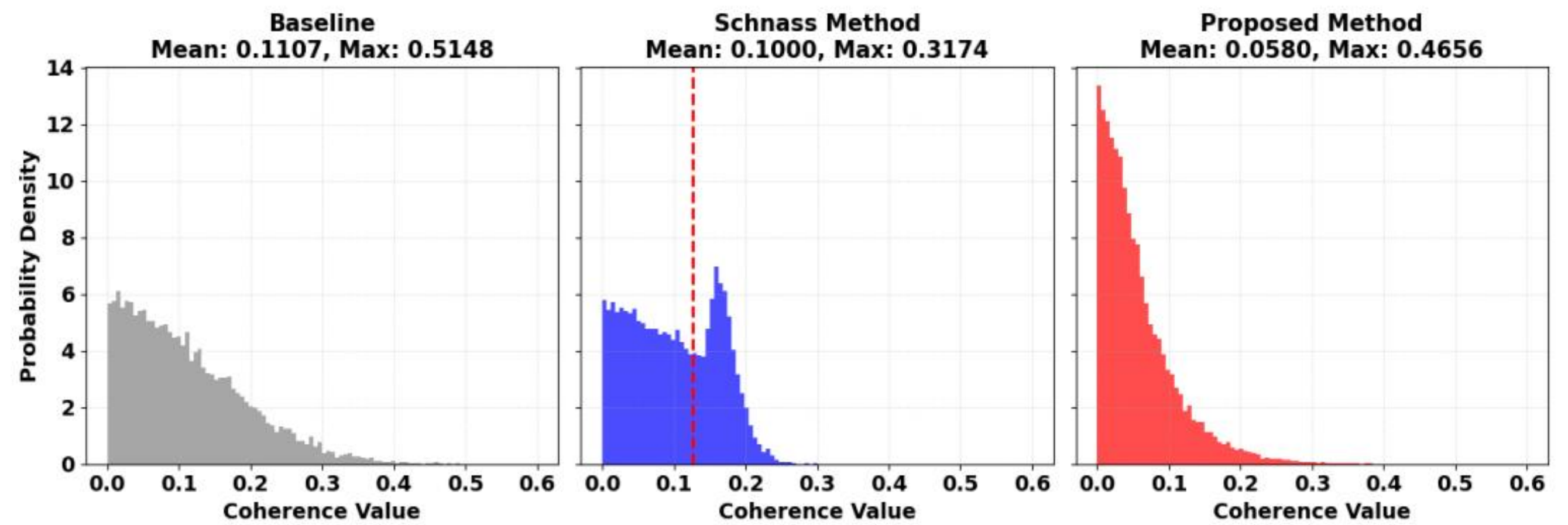


Fig. 2 Probability density histograms of effective interference correlations, with CR is 0.2, sparsity level is 30 and learning set size of 100.

Experimental Results

Accuracy & Efficiency: Under noisy conditions, the proposed method significantly improves SRR (+0.6 vs. Baseline) and NRE (-0.4 vs. Baseline) while maintaining a low computational overhead (+0.36 ms vs. Baseline) on power-law decaying sparse signals.

- Support Recovery Rate (SRR): to assess whether the correct atoms are selected.
- Normalized Reconstruction Error (NRE): to evaluate the reconstruction accuracy.

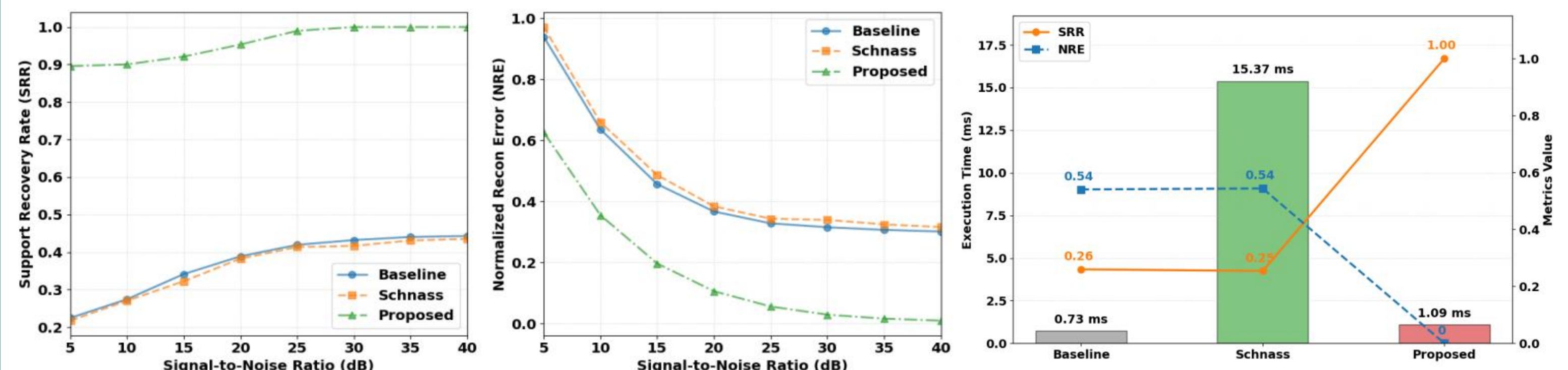


Fig. 3 The performance of the algorithm when the CR is 0.25 and the sparsity level is 30, with a learning length size of 100.

Visual Quality Comparision (Fig. 3 & 4):

- Clear Reconstruction Structure:** From the MNIST dataset, the reconstructed digit “5” exhibits sharper structural boundaries and a significantly clearer structural representation.
- Performance Variance:** However, for the real-sequence brain MRI dataset, the proposed method exists both reconstruction gains (red, max 0.61dB) and losses (blue, max -0.37 dB).

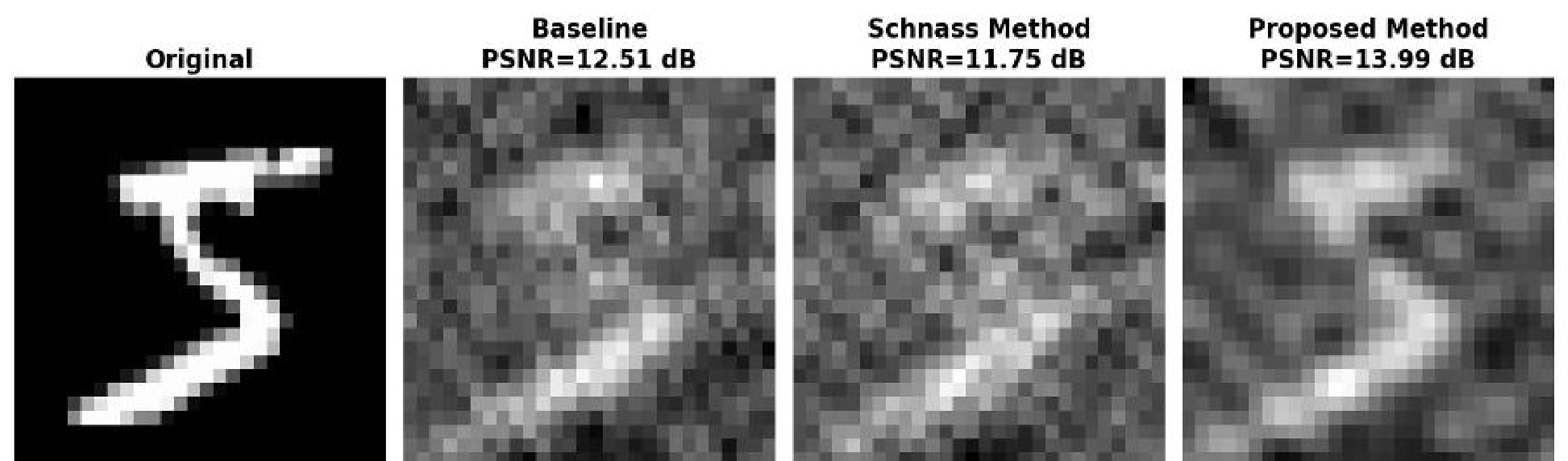


Fig. 4 Visual reconstruction comparison on the MNIST dataset with CR is 0.2, sparsity level is 30 and learning set size of 100.

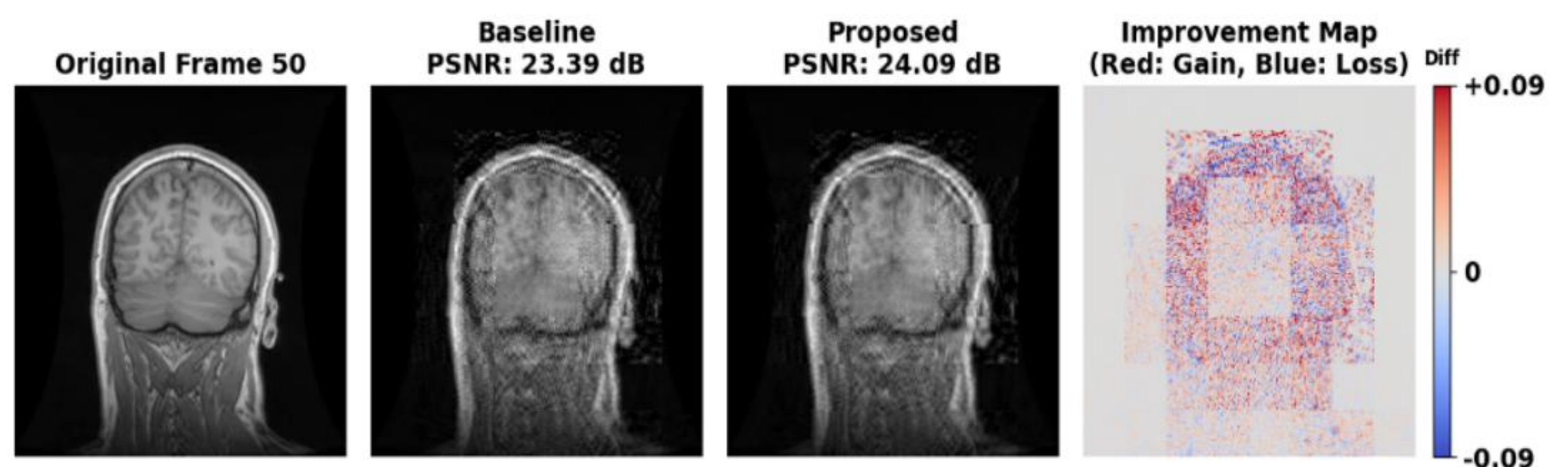


Fig. 5 Visual reconstruction comparison on the MRI sequences, with CR is 0.2 and the learning length is 10.

Conclusion and Discussion

- Conclusion:** Our proposed data-driven sensing matrix improves the atom selection and the reconstruction quality while maintaining a low computational overhead.
- Limitation:** Performance is highly sensitive to the quality and similarity of the training data.
- Future work:** 1, Integrate robust regularization and adaptive prior selection strategies. 2, Improve computational efficiency for high-dimensional signals.